

RAK12040 WisBlock 8x8 IR Sensor Datasheet

Overview

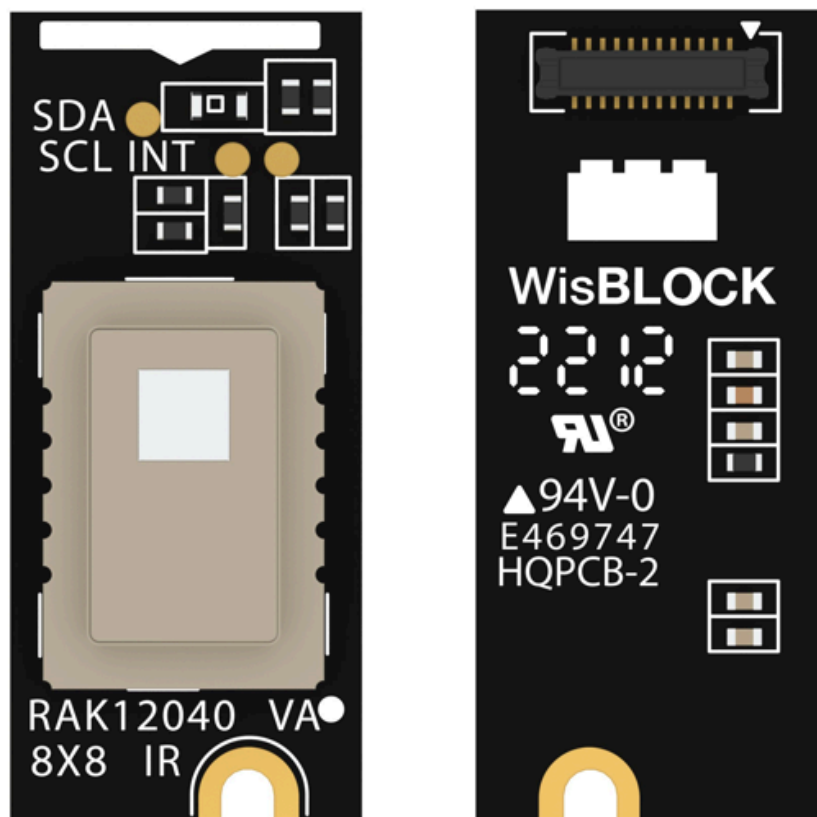


Figure 1: RAK12040 WisBlock 8x8 IR Sensor

Description

RAK12040 is a WisBlock 8x8 IR Sensor Module that extends the WisBlock system with an AMG8833 chip from Panasonic. It comes with a ready-to-use software library and tutorial, making it easy to build a temperature data acquisition system. Temperature data is interfaced via I2C. Additionally, the RAK12040 can be mounted to sensor slot of WisBlock Base board.

Product Features

- Sensor specifications

- Based on AMG8833
- Temperature detection of two-dimensional area: 8×8 (64 pixels)
- I2C digital interfaces
- 3.3 V power supply
- Operating temperature range: 0 to +80 °C
- **Size**
 - 10 × 23 mm

Specifications

Overview

Mounting

Figure 2 shows the mounting mechanism of the RAK12040 module on a [WisBlock Base](#) board. The RAK12040 module can be mounted on the slots: **A & C**.

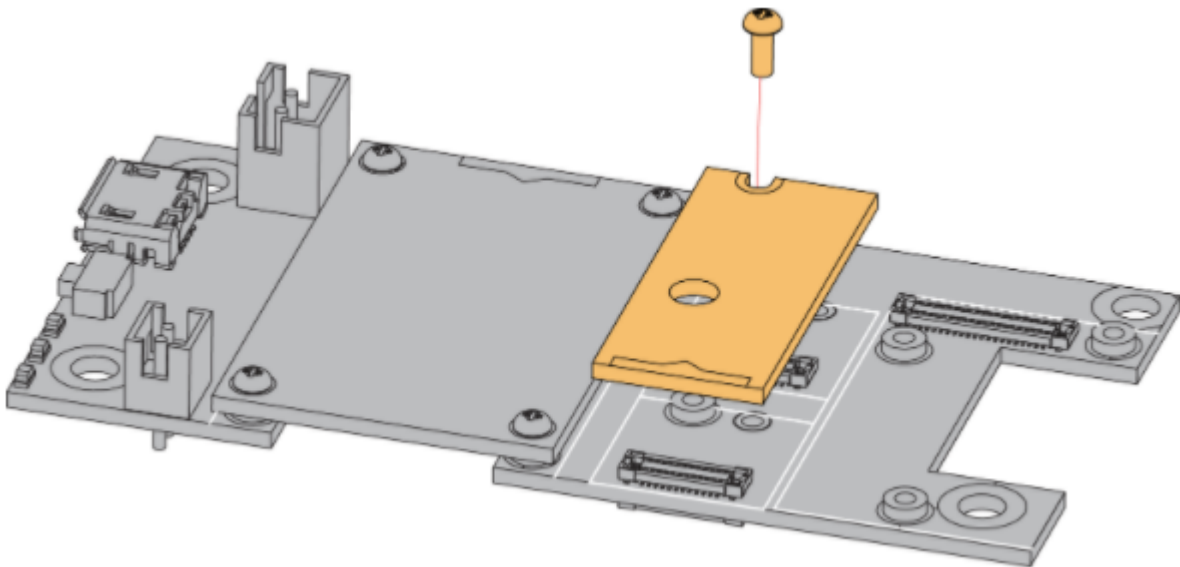


Figure 2: RAK12040 WisBlock 8×8 IR Sensor Mounting

Hardware

The hardware specification is categorized into five parts. It shows the chipset of the module and discusses the pinouts, sensors, and the corresponding functions and diagrams. It also covers the electrical and mechanical parameters, including the tabular data of the functionalities and standard values of the RAK12040 WisBlock 8×8 IR Sensor.

Chipset

Vendor	Part Number
Panasonic	AMG8833

Pin Definition

The RAK12040 WisBlock 8×8 IR Sensor comprises a standard WisBlock connector. The WisBlock connector allows the RAK12040 module to be mounted to a WisBlock Base board. The pin order of the connector and the pinout definition are shown in **Figure 3**.

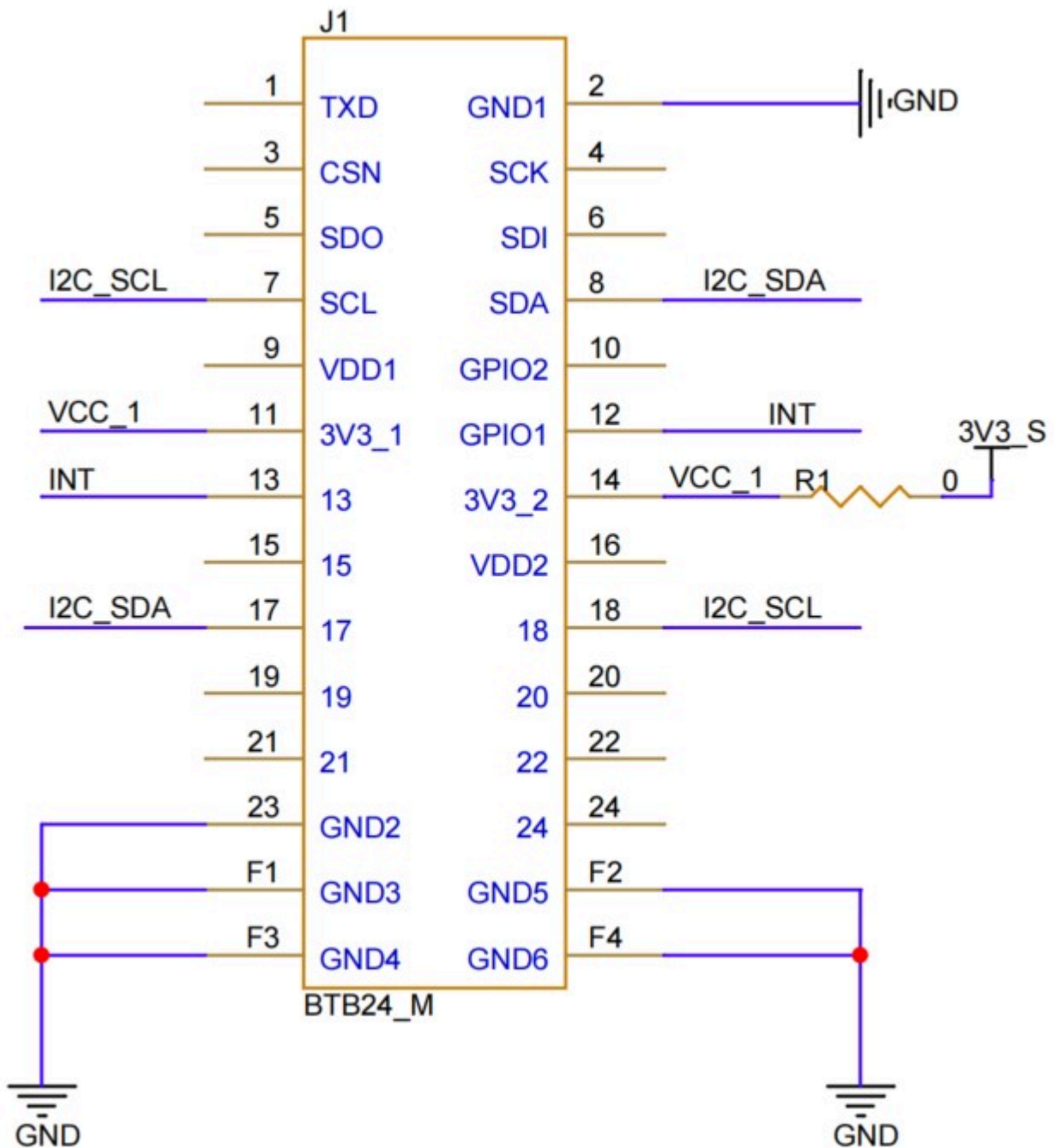


Figure 3: RAK12040 WisBlock 8x8 IR Sensor Pinout Diagram

NOTE

When the RAK12040 is used with other sensor modules powered by **3V3_S** together, it can not be mounted on slot **A**.

Electrical Characteristics

Recommended Operating Conditions

Symbol	Description	Min.	Nom.	Max.	Unit
VCC	Power Supply Voltage	-	3.3	-	V

Mechanical Characteristics

Board Dimensions

Figure 4 shows the dimensions and the mechanic drawing of the RAK12040 module.

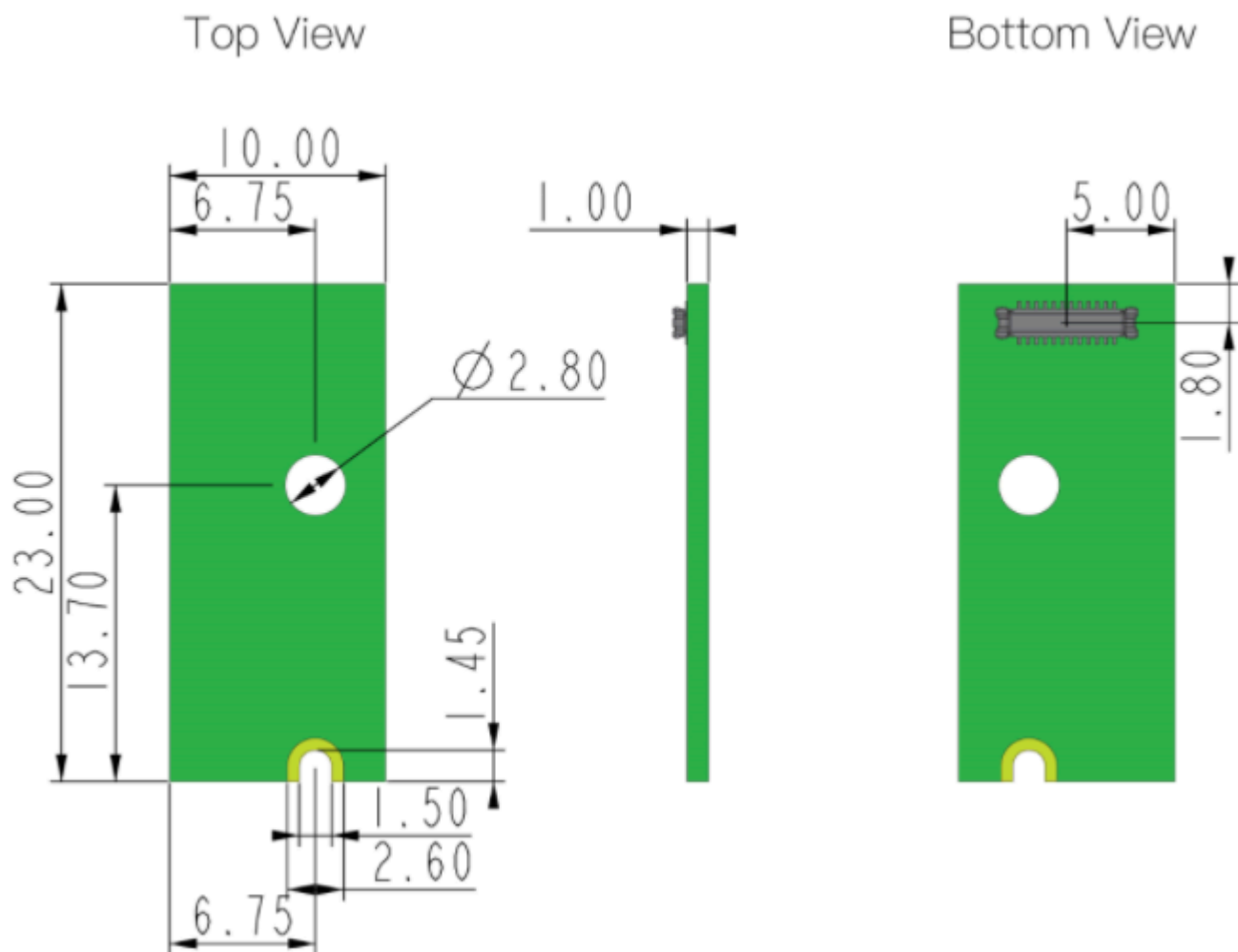


Figure 4: RAK12040 WisBlock 8x8 IR Sensor Mechanic Drawing

WisConnector PCB Layout

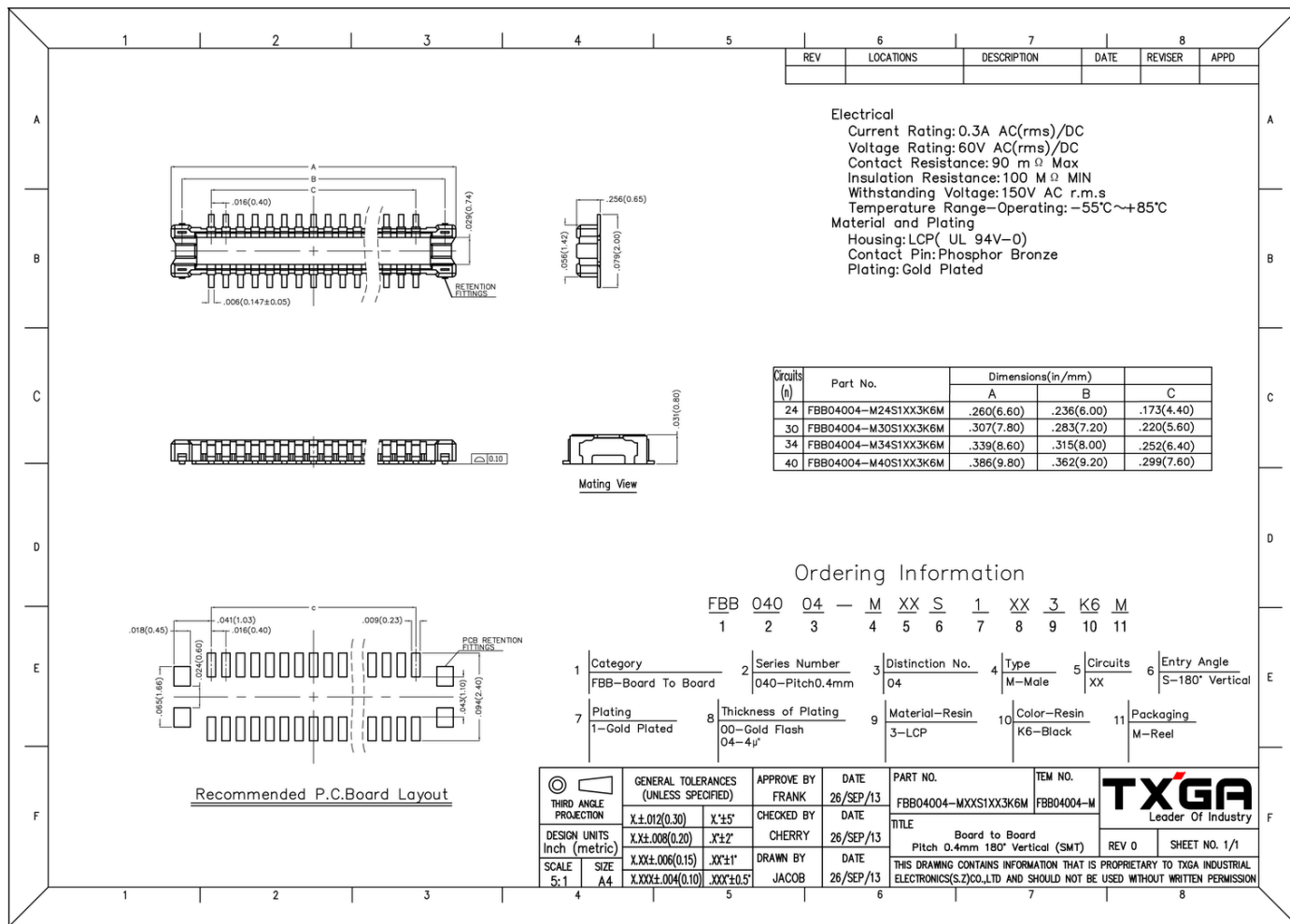


Figure 5: WisConnector PCB footprint and recommendations

Schematic Diagram

Figure 6 shows the schematic of the RAK12040 module.

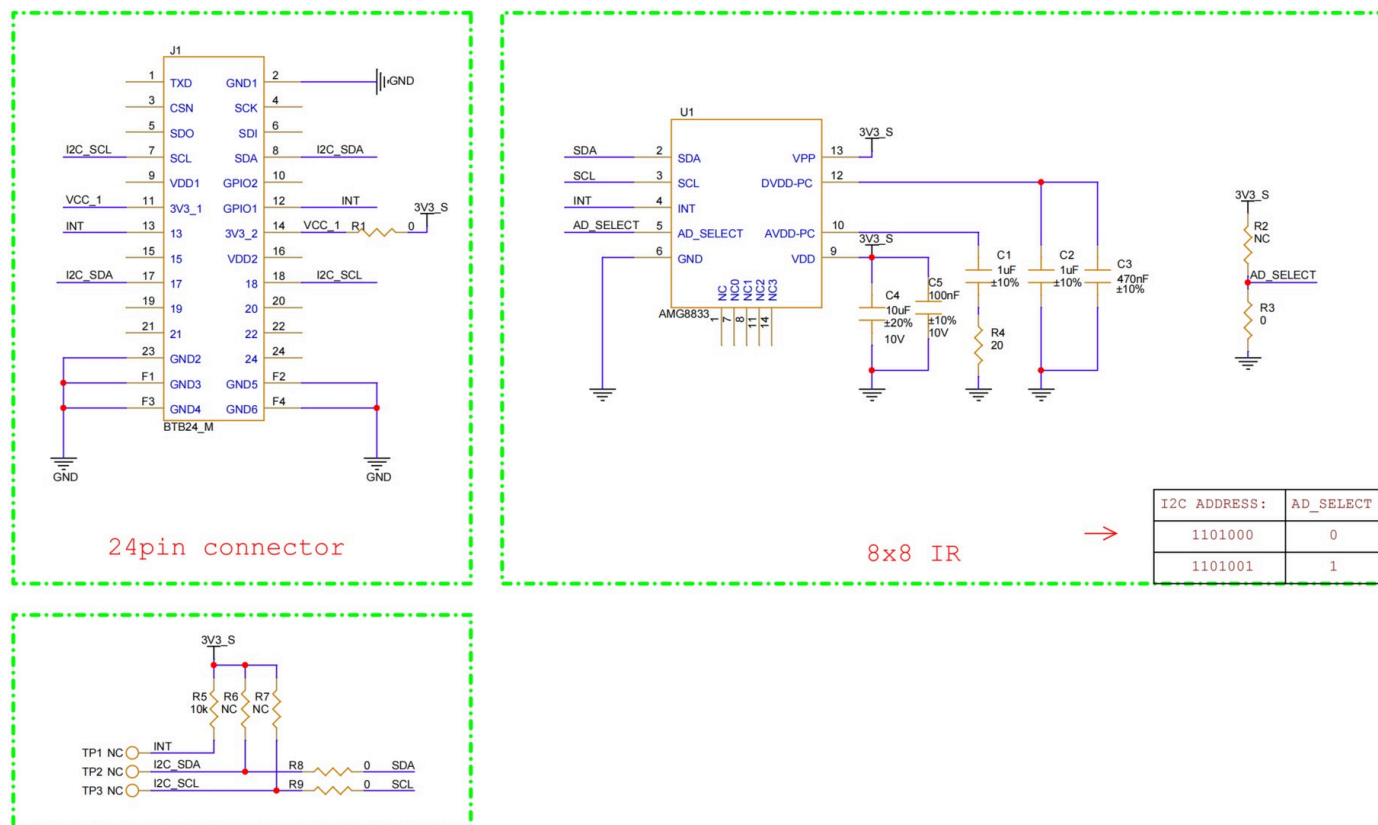


Figure 6: RAK12040 WisBlock 8x8 IR Sensor schematics